Paleomagnetism and *K-Ar* Ages of Successive Lava Flows* (2)

——Kita-Matsuura basalt, Kyushu, Japan——

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Abstract

Paleomagnetism and *K-Ar* ages of two successive lava flows from Kita-Matsuura, northwestern Kyushu were studied in attempting to estimate the time required for the geomagnetic field to switch from one polarity to the other. Both lava flow successions record normal and reversed polarities. *K-Ar* ages of these flows range from 7 m.y. to 10 m.y. However, because of inconsistensies within a flow and among successive flows, *K-Ar* ages should be regarded as indicating approximate ages of these lava flows. Hence, it was not possible to determine the transition time for the geomagnetic field switching.

1. Introduction

There remains now little doubt that the geomagnetic field has changed polarity at least a few tens of times during the last ten million years (*Cox, et al*, 1965, *McDougall, I., et al*, 1966). The transition time for changing from one polarity to the other is then essential for an understanding of the origin of the geomagnetic field. In addition, this information would yield an important approach for estimating the electrical conductivity in the core (*Verhoogen*, 1966).

Nincovich et al (1966) from paleomagnetic study on deep sea sediments, estimated the time interval from sedimentation rates and suggested that the transition time was of the order of 10^3–10^4 years. *Cox* and *Dalrymple* (1967) obtained about 4600 years for the transition time on the basis of statistical considerations of the overlaps of *K-Ar* ages of normal and reversed polarity sections in late Tertiary and Quaternary lavas. Because of some uncertainty in the estimate of the sedimentation rate, the first estimate can not be regarded as conclusive. The second estimate yields only an average time interval of various transitions, but there is no reason to expect that transition time intervals were the same for all transitions throughout geological time. Direct application of *K-Ar* dating on successive lava flows which recorded both polarities may then provide a third approach to the above problem.

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On the basis of *K-Ar* ages of successive lava flows from New Mexico, *Ozima et al* (1967) suggested about 0.05 m.y. as a maximum time interval for the transition 3.6 m.y. ago. The present paper is the second one in the series of studies on *K-Ar* ages and paleomagnetism of successive lava flows which recorded both polarities. As lava eruption is intermittent, direct application of *K-Ar* dating to such lava flow sequences yields only upper estimate for the transition time. In addition, because of limitted experimental precision in *K-Ar* dating, the time interval which can be resolved by this method becomes progressively larger for older ages (*Cox* and *Dalrymple*, 1967). In spite of these difficulties, however, this approach may be useful in resolving the problem of whether the time required for the geomagnetic polarity switching was different for different transition, or has remained essentially constant.

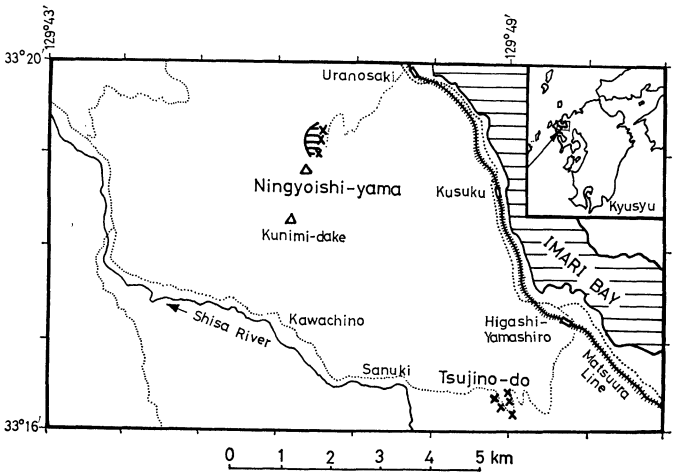


Fig. 1 Geological map of sampling sites.

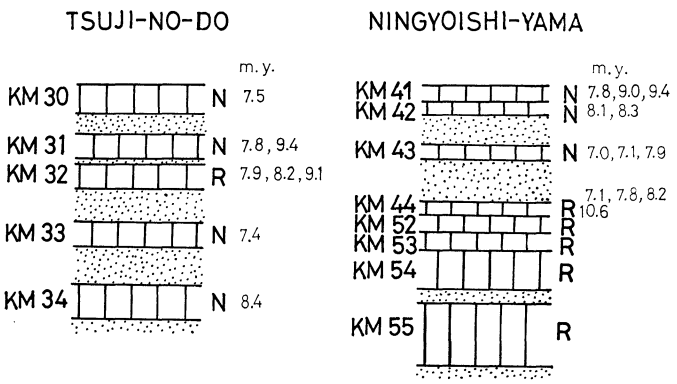


Fig. 2 Schematic representation of lava flow successions at the Tsujinodo and Ningyoishi-Yama sites.
N and *R* indicate normal and reversed polarities in each flow.
K-Ar ages (in million years) are also shown.

2. Geological description

In northwestern Kyushu, volcanic rocks are distributed approximately over an area of 250 km^2 (Figure 1). These are named the Kita-Matsuura basalts. The principal member of the volcanic rocks is alkaline basalt. On the basis of stratigraphic observation and paleomagnetic correlation, *Kurasawa* (1967) concluded that the eruption of the lavas had occured over a rather long period ranging from middle Pliocene to early Pleistocene.

At Tsujinodo and Ningyoishi-yama, successive lava flows are exposed with interbedded sediments and volcanic breccias. These are schematically illustrated in Figure 2. At the Tsujinodo site, there are five successive lava flows and at the Ningyoishi-yama site eight successive lava flows are observed. The stratigraphic relation between the two sites is not clear.

3. Paleomagnetic results

A total of 25 hand samples were obtained from five successive lava flows at Tsujinodo site and 40 from eight successive lava flows at Ningyoishi-yama site. One or two cores are drilled from each hand sample. All core samples were subjected to *a-c* demagnetization test with peak field of 150 *Oe*. For most of the samples the NRM directions did not change on *a-c* demagnetization. In table 1 the results of the paleomagnetic measurements, including NRM directions before and after *a-c* demagnetization, dispersion factors etc, are listed. In Figure 3 virtual poles after *a-c* demagnetization are shown for the Tsujinodo and Ningyoishi-yama sites.

At the Tsujinodo site, the upper two flows and the lower two flows (KM 30, 31, 33, 34) have a normal polarity, whereas the middle flow (KM 32) has a reversed direction.

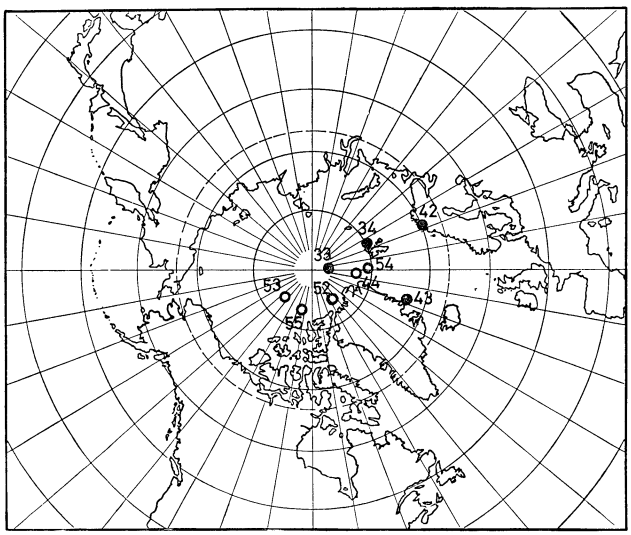


Fig. 3 Equal area projection of paleomagnetic virtual pole positions of lava flows after *a-c* demagnetization with a peak field of $H=150$ *Oe*. Solid circles represent normal polarity; open circles represent reversed polarity.

However, *a-c* demagnetization of the reversed flow with a peak field of 150 *Oe* resulted in a change in the NRM directions of more than 50 degrees. Grouping of NRMs of these samples before and after *a-c* demagnetization is also very poor as shown in Table 1. Hence, it is doubtful that the NRMs of these samples represent a reversed geomagnetic field.

Except for flow 32, however, the NRMs show a good grouping and also good stability against *a-c* demagnetization (Table 1). Hence, we conclude that the measured NRM directions represent the paleogeomagnetic field. At both sites, normal NRM directions have almost the same directions as that of the present geomagnetic field, whereas the reversed one has almost the opposite direction.

Table 1.
Tsujinodo site

Sample	*a-c* demagnetization*	*N*	NRM				Virtual pole		Intensity (emu/g)
			I	*D*	*k*	α_{95}	lat	long	
KM30	$\tilde{H}=0$	5	72.5	47.0	8.03	28.8	50.2	166.8	2.11×10^{-4}
	$\tilde{H}=150\,Oe$	5	42.2	122.7	9.5	26.1	−10.6	180.5	8.30×10^{-5}
KM31	$\tilde{H}=0$	5	68.6	0.0	27.6	14.8	71.3	129.4	3.21×10^{-4}
	$\tilde{H}=150\,Oe$	5	65.9	57.3	15.4	20.1	45.2	182.1	1.82×10^{-4}
KM32	$\tilde{H}=0$	5	1.2	−149.4	5.5	36.0	−45.5	82.6	2.70×10^{-4}
	$\tilde{H}=150\,Oe$	5	7.8	164.3	1.2	142.5	−49.9	154.1	1.24×10^{-4}
KM33	$\tilde{H}=0$	5	47.3	−2.5	183.4	5.7	84.7	334.4	9.42×10^{-4}
	$\tilde{H}=150\,Oe$	5	51.3	−1.9	58.2	10.1	87.9	1.3	5.29×10^{-4}
KM34	$\tilde{H}=0$	5	48.3	−19.1	173.6	5.8	73.3	30.8	1.38×10^{-3}
	$\tilde{H}=150\,Oe$	5	49.8	−12.4	184.6	5.6	79.1	28.6	1.07×10^{-3}
Ningyoishi-yama site									
KM41	$\tilde{H}=0$	4	52.1	−146.5	4.29	50.5	−16.9	100.3	1.55×10^{-4}
	$\tilde{H}=150\,Oe$	4	35.0	−143.7	3.7	55.4	−27.0	90.4	9.30×10^{-5}
KM42	$\tilde{H}=0$	4	42.4	−21.6	66.5	11.3	69.3	20.0	1.35×10^{-3}
	$\tilde{H}=150\,Oe$	4	43.3	−21.4	80.8	10.3	69.7	21.6	1.12×10^{-3}
KM43	$\tilde{H}=0$	5	38.8	−13.2	59.9	10.0	73.8	358.4	7.50×10^{-4}
	$\tilde{H}=150\,Oe$	5	36.8	−9.1	39.5	12.3	74.9	344.0	4.22×10^{-4}
KM44	$\tilde{H}=0$	8	−46.3	173.9	52.1	6.9	−82.5	173.5	1.36×10^{-3}
	$\tilde{H}=150\,Oe$	8	−46.6	174.5	67.8	6.0	−82.7	173.3	1.25×10^{-3}
KM52	$\tilde{H}=0$	4	−46.0	179.9	127.1	8.2	−84.1	130.4	1.49×10^{-3}
	$\tilde{H}=150\,Oe$	4	−46.9	−179.5	127.6	8.2	−84.8	124.4	1.28×10^{-3}
KM53	$\tilde{H}=0$	5	−50.1	−173.0	210.2	5.3	−83.6	59.4	1.75×10^{-3}
	$\tilde{H}=150\,Oe$	5	−50.4	−172.3	212.7	5.3	−83.1	55.4	1.59×10^{-3}
KM54	$\tilde{H}=0$	5	−44.0	170.6	53.5	10.6	−78.9	179.4	2.40×10^{-3}
	$\tilde{H}=150\,Oe$	5	−46.0	171.8	66.0	9.5	−80.8	181.7	1.99×10^{-3}
KM55	$\tilde{H}=0$	5	−44.7	−173.6	55.2	10.4	−81.1	89.2	1.80×10^{-3}
	$\tilde{H}=150\,Oe$	5	−47.0	−172.9	75.0	8.9	−82.1	77.2	1.33×10^{-3}

*, $H=0$ denotes an original NRM. $H=150\,Oe$ denotes the NRM after *a-c* demagnetization with $H=150\,Oe$. *N*, number of samples. *I*, Inclinations of mean direction of NRM. *D*, Declinations of mean direction of NRM. *k*, Fisher's precision parameter. α_{95}, Semiangle of 95% confidence for mean direction.

4. *K-Ar* ages of lava flows

Prior to *K-Ar* dating, thin sections of all samples were examined. The petrographic descriptions are given in the appendix. Slight alteration of minerals and the presence of a few glassy parts are common in all samples. *K*-analysis was done by a flame photometer with Li internal standard and *Ar* analysis by the isotope dilution method with a 15 cm radius pyrex glass mass spectrometer. In *Ar* analysis a static operation was used. Further detailed descriptions of experimental procedures are given elsewhere (*Ozima et al*, 1967).

Results of *K-Ar* datings were given in Table 2. In general ages for all lava flows cluster around 8–9 m.y. However, ages are generally not consistent within a single flow and among successive flows: except flows 4202 and 4303-b, *K-Ar* ages of samples from the same flow do not agree with each other and some of the upper flows show older ages than those of the underlying flows (for example, flow 4101-b, 3104-b). There should be no significant difference in analytical precision between the present analyses and those in the New Mexico samples reported previously, in which *K-Ar* ages are generally consistent within five percent (*Ozima et al*, 1967). In addition, air contamination in the present analyses are generally lower than in the case of New Mexico samples. Hence, it is unlikely that the observed

Table 2.
Tsujinodo-site

Sample	wt (g)	Ar^{40}_{rad} (mole/gr)	$\frac{(Ar^{40})\ \text{air}}{(Ar^{40})\ \text{total}} \times 100$	K (%)	t (m.y.)
KM3001-a	8.553	1.712×10^{-11}	72.5	1.28	7.52
KM3104-a	7.782	2.050×10^{-11}	56.2	1.47	7.82
KM3104-b	7.788	2.459×10^{-11}	45.7	1.46	9.44
KM3203-a	11.32	1.516×10^{-11}	63.7	1.08	7.87
KM3203-b	7.033	1.740×10^{-11}	84.0	1.07	9.12
KM3203-b	5.934	1.566×10^{-11}	81.6	1.07	8.20
KM3301-b	8.033	6.717×10^{-12}	95.2	0.512	7.35
KM3403-a	7.227	8.136×10^{-12}	91.2	0.543	8.40
Ningyoishi-yama site					
KM4101-a	10.74	2.370×10^{-11}	51.0	1.71	7.77
KM4101-b	11.327	2.732×10^{-11}	81.5	1.63	9.39
KM4101-b	9.523	2.620×10^{-11}	87.3	1.63	9.01
KM4202-c1	7.944	3.314×10^{-11}	70.6	2.29	8.11
KM4202-c2	9.684	3.361×10^{-11}	79.5	2.27	8.30
KM4303-a	7.076	4.548×10^{-11}	42.4	2.23	7.92
KM4303-b	6.086	4.044×10^{-11}	74.6	3.22	7.04
KM4303-b	6.871	4.136×10^{-11}	40.2	3.26	7.11
KM4403-a	8.029	1.502×10^{-11}	91.6	0.791	10.6
KM4403-b	7.381	1.037×10^{-11}	84.1	0.824	7.05
KM4403-b	8.306	1.202×10^{-11}	79.0	0.824	8.18
KM4404-a	9.486	1.087×10^{-11}	81.0	0.777	7.84

$\lambda_e=0.585\times10^{-10}y^{-1}$, $\lambda_\beta=4.72\times10^{-10}y^{-1}$, $K^{40}/K=1.19\times10^{-4}$ mole/mole

inconsistencies in the age results are merely due to analytical errors. The inconsistent ages in the present analyses must be due to alteration of samples and/or to the presence of glassy parts.

Consequently all ages should be considered as giving only a rough measure of the ages of the flows, even though flows 4202 and 4303-b give a good reproducibility of the *K-Ar* ages within the flows.

5. Discussion and conclusions

K-Ar ages obtained for four samples at Tsujinodo site and five samples at Ningyoishi-yama site range from 7 m.y. to 10 m.y, the average age being about 8.0 m.y. It can be concluded that the Kita-Matsuura basalts were erupted 8.0 ± 1.0 m.y. ago.

However, because of the inconsistencies among ages obtained from successive lava flows, it was not possible to resolve the time interval between two successive flows. Hence, the transition time from one magnetic polarity to the other could not be estimated. The inconsistent ages are probably due to alteration of samples.

To obtain reliable *K-Ar* ages, it is customary to examine thin sections of each sample prior to *Ar* analysis (*Dalrymple*, 1963, *McDougall*, 1964). However, as there is no objective standard to judge the suitability of samples for *K-Ar* dating, there is still some ambiguity as to reliability of *K-Ar* ages even after the severest choice of samples under a microscope. Realizing that for young basaltic rocks, there is no other available dating method which can be used to check *K-Ar* ages, it is particulary important to establish a more objective criterion of the reliability of *K-Ar* ages. Internal consistensies within a single flow and among flows whose succession are geologically well established may be one of the most practical and objective criteria of the reliability of *K-Ar* ages.

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Appendix

Petrographic description of lava flows

KM3001: Alkali olivine basalt

Feldspars (plagioclase, alkali feldspar) are abundunt (60–70%) and a little altered. Pyroxene and olivine are observed in groundmass and seem to be altered. Opaque minerals are abundant (10–15%). Glassy parts are few (~5%). No xenolith or xenocryst are observed.

KM3104: Alkali olivine basalt

Feldspars (plagioclase, alkali feldspar) are altered. Olivine is fairly altered to iddingsite. Pyroxene is observed in groundmass and altered. Some clay minerals are observed. Glassy parts are few (~5%). There are no xenolith or xenocryst.

KM3203: Alkali olivine basalt

Feldspars (plagioclase, alkali feldspar, 60–70%) are fairly altered. Olivine (~12%) is altered. Pyroxene is comparatively few. Chlorite is observed. Glassy parts are about 10% and opaque minerals are about 5%. Xenocrysts of plagioclase are observed.

KM3301: Alkali olivine basalt

Feldspars (plagioclase, alkali feldspar) are about 40–50%. Olivine is a little altered. In groundmass, pyroxene and some clay minerals are observed. Glassy parts are very few (~5%). Opaque minerals are about 5%. No xenolith or xenocryst are observed.

KM3403: Alkali olivine basalt

Feldspars (alkali feldspar, plagioclase, ~50%) are a little altered. Olivine is partly altered. Secondary minerals such as chlorite are observed. Glassy parts are about 10%. Opaque minerals are few (~5%). No xenolith or xenocryst are observed.

KM4101-a and KM 4101-b: Trachybasalt (or alkali-olivine basalt)

Trachytic texture. Many crystals of olivine are observed (20–30%) and partly altered. Feldspars (plagioclase, alkali feldspar) are slightly altered. Alkali feldspar is comparatively abundant. Pyroxene is observed. Glassy parts are very few (~5%, almost holocrystalline). No xenolith or xenocryst are observed. Opaque minerals are few (~5%).

KM 4202-c: Alkali olivine basalt

Grain size of phenocrysts is comparatively fine. Olivine is partly altered. Feldspars (plagioclase, alkali feldspar ~60%,) are slightly altered. Pyroxene is observed in groundmass. Glassy parts are also observed (10–20%). Opaque minerals are few (~5%). Xenolith and xenocrysts are not observed.

KM-4303-a and KM 4303-b: Alkali olivine basalt

Olivine is altered to iddingsite. Feldspars (plagioclase, alkali feldspar) are a little altered. Pyroxene is a little altered. Glassy parts are observed (10–20%) and divitrified. Xenolith and xenocryst are not observed.

KM4403: Alkali olivine basalt

Feldspars (plagioclase, alkali feldspar) are abundant (~60%). Olivine is abundant (~20%) and partly altered to iddingsite. Pyroxene is a little altered. Glassy parts are few (~5%). Opaque minerals are few (~5%). No xenolith or xenocryst are observed.